PRIME MINISTER'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

ARUSHA REGION

FORM TWO PRE - MOCK EXAMINATION

031 **PHYSICS**

Time: 2:30 Hours MARCH 2026

INSTRUCTIONS

1. This paper consists of section A, B and C with total of ten (10) questions.
2. Answer all questions in all sections
3. Read carefully the instructions given in each section
4. Write your examination number on every page
5. Where necessary the following constat my be used
   (i) Acceleration due to gravity of = 10N/kg = $10m/s^2$
   (ii) Density of water = $1g/cm^3$ = $1000Kg/ m^3$

NO.	SCORE	SIGNATURE
1.		
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10.		
TOTAL		
CHECKER'S SIGN		

SECTION A (15 marks)

Answer all questions in this section

1. For each of the items (i) – (x), choose the correct answer among the given alternatives and write its letter beside the item number in the answer sheet provided.

(i) Juma wants to measure the diameter of piece of wire which instrument can enable him to get accurate reading.
(A) Vernier caliper
(B) Rate meter ruler
(C) Micrometer screw gauge
(D) Engineers calliper

(ii) Why is hot soup testier than cold soup?
(A) It has higher surface tension
(B) It has lower surface tension
(C) It has higher capillary action
(D) It has lower capillary action

(iii) A force acting on an object cause some properties of the object to change. Which of the list contains properties that can be changed by the action of force
(A) Mass, motion, and shapes
(B) Mass, motion and size
(C) Mass, shape and size
(D) Motion, shape and size

(iv) Which of the following is a pair of vectors quantities
(A) Force and velocity
(B) Speed and mass
(C) Energy and displacement
(D) Time and temperature

(v) The upward force exerted by a liquid on a submerged object is called
(A) Buogent force
(B) Gravity

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(C) Friction
(D) Momentum

(vi) The work done when a force of 20N moves an object 5m in the direction of the force is
(A) 4J (B) 100J (C) 25J (D) 10J

(vii) If a body is moving with constant velocity it's acceleration is
(A) Decreasing
(B) Zero
(C) Constant but not zero
(D) Increasing

(viii) The food consumed by human being produces
(A) Only kinetic energy
(B) Only heat energy
(C) Both heat energy and chemical energy
(D) Only chemical energy

(ix) Which of the following are the non-magnetic metal
(A) Iron and steel
(B) Aluminum and zinc
(C) Nickel and cobalt
(D) Nickle and stool

(x) A ship sink lower in fresh water than in sea water, what can you conclude about the density:-
(A) Fresh water is denser than the sea water
(B) The density of sea water is the same as the ship
(C) The sea water is denser than fresh water
(D) Sea water and fresh water have the same density

Qns	i	ii	iii	iv	v	vi	vii	viii	ix	x
Answ										

2. Match the items in List A with the response in List B by writing the letter of the correct response beside the item number in the space provided.

LIST A	LIST B
(i) A force that explains difficult in pouring honey out of the vessel.	A. Torsional force B. Tension force C. Gravity

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(ii) A force of attraction between object and the earth (iii) A force that causes object to float (iv) A force transmitted through a string when pulled tightly. (v) A force that opposes relative motion between two solid bodies is constant	D. Frictional force E. Bouget force F. Viscous force G. Compressional force H. Normal force

SECTION B (70 marks)

Answer all questions

3. (a) Define the following terms

(i) Magnetization

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(ii) Demagnetization

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(iii) Magnetic field

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(b)List three properties of magnet

(i) …….
(ii) …….
(iii)……

(c) Mention three (3) types of magnets according to their shapes

(i) ……..

(ii)…….

(iii)……

4. (a) Outline three applications of density and relative density in real life.

(i) ……
(ii) ……
(iii) ……

(b)An object p of mass 0.5kg. When it is totally immersed in the water of volume $55cm^3$

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the final water volume is 65cm^3. Calculate:

(i) The volume of the object P.

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(ii) The density of the object P.

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(c)Calculate the upthrust force experienced by a steel ball of radius 8cm when immersed in water.

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5. (a) As a form two students who is expert in physics give the difference between Kinetic energy and potential energy.

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(b)A ball of mass 200g fall freely from a height of 20m above the ground and hits a concrete floor and rebounds to a height of 5m; Find the;

(i) P. E of the ball before it fall.

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(ii) Its K. E as it hits the concrete.

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(iii)Velocity with which it hits the concrete

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(iv)K. E as it rebounds

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6. (a) Briefly explain why
   (i) Water wets glass but mercury does not
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   (ii) Mosquitoes can manage to walk on the surface of water.
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   (b) A spring is stretched by 0.5m by a weight of 5N hung from one end.

   (i) What wight will stretch it by 0.03m?
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   (ii) Determine the spring constant
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7. (a) Why a tractor with wide typer cannot get stuck in muddy places compared to a car with narrow tires.
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   (b) List three factors which the pressure in liquid depends on.
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   (c)Calculate the area of a surface of an object which exerts pressure of $0.2N/m^2$ when the force acting on it is 2N.
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8. (a) Define the following terms as used in concave mirror.

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(i) Principal focus

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(ii) Center of curvature

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(b)The focal length of a concave mirror is 4cm. An object 1.5cm high is placed 12cm in front of the mirror. Determine.

(i) Position of image

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(ii) Size of image

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(iii) Nature of the image

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9. (a)Define uniform velocity as applied in straight line.

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(b)Some of the form two students confuse to differentiate between equations of linear motion in straight line and the equations of motion under gravity. Use three equations of linear motion to derive three equations of motion under gravity.

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(e)A car starts from rest and accelerates uniformly at a rate of $4M/S^2$ for 5 seconds. It maintain a constant velocity for 20 seconds. The brakes are then applied and the car decelerates for 3 seconds.

(i) Draw the velocity time graph to represent the motion. Hence calculate:-

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(ii) The maximum velocity attained

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(iii) The total distance covers.

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SECTION C (15 marks)

Answer question ten (10)

10. (a) Three resistors of 4Ω, 6Ω and 12Ω are connected in parallel.

(i) Draw their circuit diagram

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(ii) Calculate the total resistance

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(b)(i) Three capacitors of 3μF, 10μF and 15μF are connected in paralled. Find the total capacitance and state one (1) advantage of connecting capacitors in parallel.

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(iii) Define lighting

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(c) Suppose you are physics laboratory leader and you are asked by your teacher to prepare the electrical components and instruments for an experiment to determine the relationship between voltage and current list down electrical components which will be used in this experiment.

(i) ……
(ii) ……
(iii) ……
(iv) ……
(v) ……

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